

Smart Scan Strategy for Electronic Warfare

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Class:**B.SC.SYIT**

Project Title:**Smart Scan Strategy for Electronic Warfare**

1. Project Description:

Smart Scan Strategy for Electronic Warfare is a simulation-based system that generates synthetic signal data, analyzes signal strength and confidence, and assigns HIGH, MEDIUM, or LOW priority. It uses Python Flask, REST API, SQLite, and Priority Queue to display and manage the results through a simple dashboard.

2. Code Snippet:

```
# app.py
from flask import Flask,render_template,jsonify
import random

app=Flask(__name__)

@app.route("/")
def login():
    return render_template("login.html")

@app.route("/api/scan")
def scan():
    data=[]
```

```

    for i in range(5):
        s=random.randint(20,100)
        c=random.randint(20,100)
        score=(s+c)/2
        p="HIGH" if score>=75 else "MEDIUM" if score>=50 else "LOW"
        data.append({"name":f"Signal {i+1}", "strength":s,
                    "confidence":c,"priority":p})
    return jsonify(data)

@app.route("/dashboard")
def dashboard():
    return render_template("dashboard.html")

app.run(debug=True)

```

requirements.txt

Flask

<!-- login.html -->

```

<form action="/dashboard">
<input placeholder="Email">
<input type="password" placeholder="Password">
<button>Login</button>
</form>

```

<!-- register.html -->

```

<h2>Register</h2>
<input placeholder="Email">
<input type="password" placeholder="Password">
<button>Register</button>

```

<!-- dashboard.html -->

```

<h1>SmartScan Dashboard</h1>
<button onclick="scan()">Start Scan</button>
<div id="data"></div>

```

```

<script>
function scan(){
    fetch("/api/scan").then(r=>r.json()).then(d=>{
    document.getElementById("data").innerHTML=
    d.map(x=>`${x.name} - ${x.priority}`).join("<br>");
    });
}
</script>

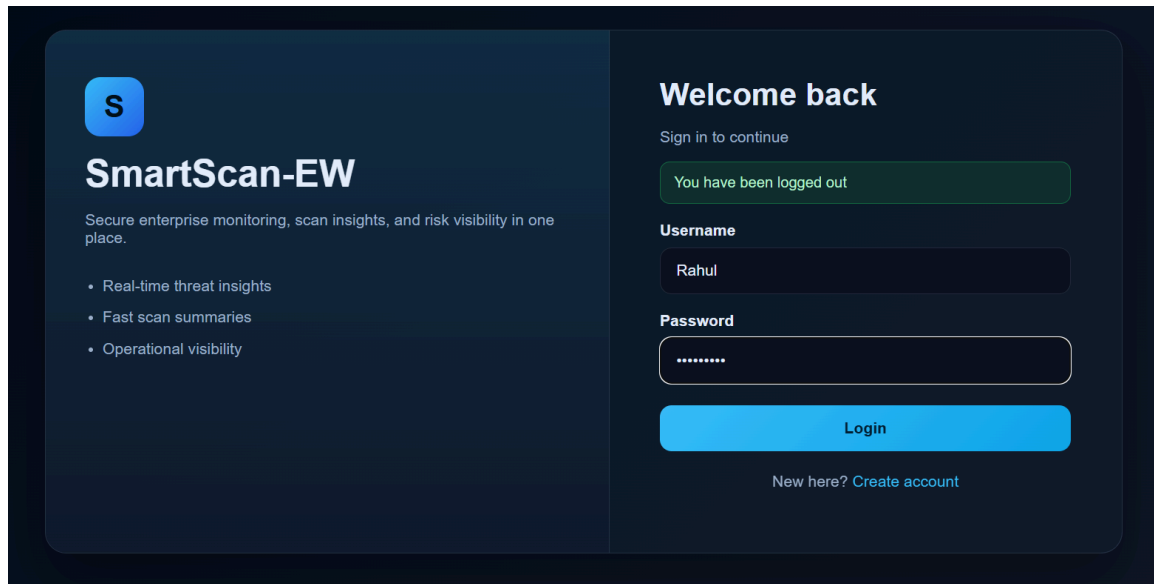
```

```
/* style.css */
```

```
body{text-align:center;font-family:Arial;padding:40px}
```

```
button,input{padding:10px;margin:5px}
```

3. Output Screenshot:



4. Video Demonstration:

<https://drive.google.com/file/d/1909o3Ct2HrNLwPAXkDD5B7rN4isyMY-C/view?usp=sharing>

5. GitHub Repository:

<https://github.com/ityadavrahul2526-cmyk/smartscan-ew.git>